Architecture & Model Relationships Guide: Merchandise Management Module

1. Executive Summary & Purpose of Merchandise Management

In a Multi-Branch Textile & Retail ERP System, the **Merchandise Management (Merchandise Planning)** module serves as the strategic & financial forecasting layer for product inventory acquisition and revenue target tracking.

Core Business Objectives:

1. **Seasonal Budget Allocation:** Establishing planned procurement budgets (`planned_budget`) for specific retail seasons (e.g., *Diwali Festival*, *Summer Casuals*, *Pongal Special*, *Wedding Silk*) to prevent over-purchasing by procurement managers.
2. **Sales Revenue Targeting:** Setting macro-level revenue targets (`target_sales`) per financial calendar year and season to evaluate store performance.
3. **Inventory & Assortment Control:** Aligning product catalog taxonomies (Categories, Sub-Categories, Brands, Textile Attributes, and Price Lists) with seasonal consumer demand.

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2. Model Relationship Architecture

2.1 Direct Relationships in `MerchandisePlanning` Model

The `MerchandisePlanning` model connects macro-level financial goals to specific time periods and retail seasons:

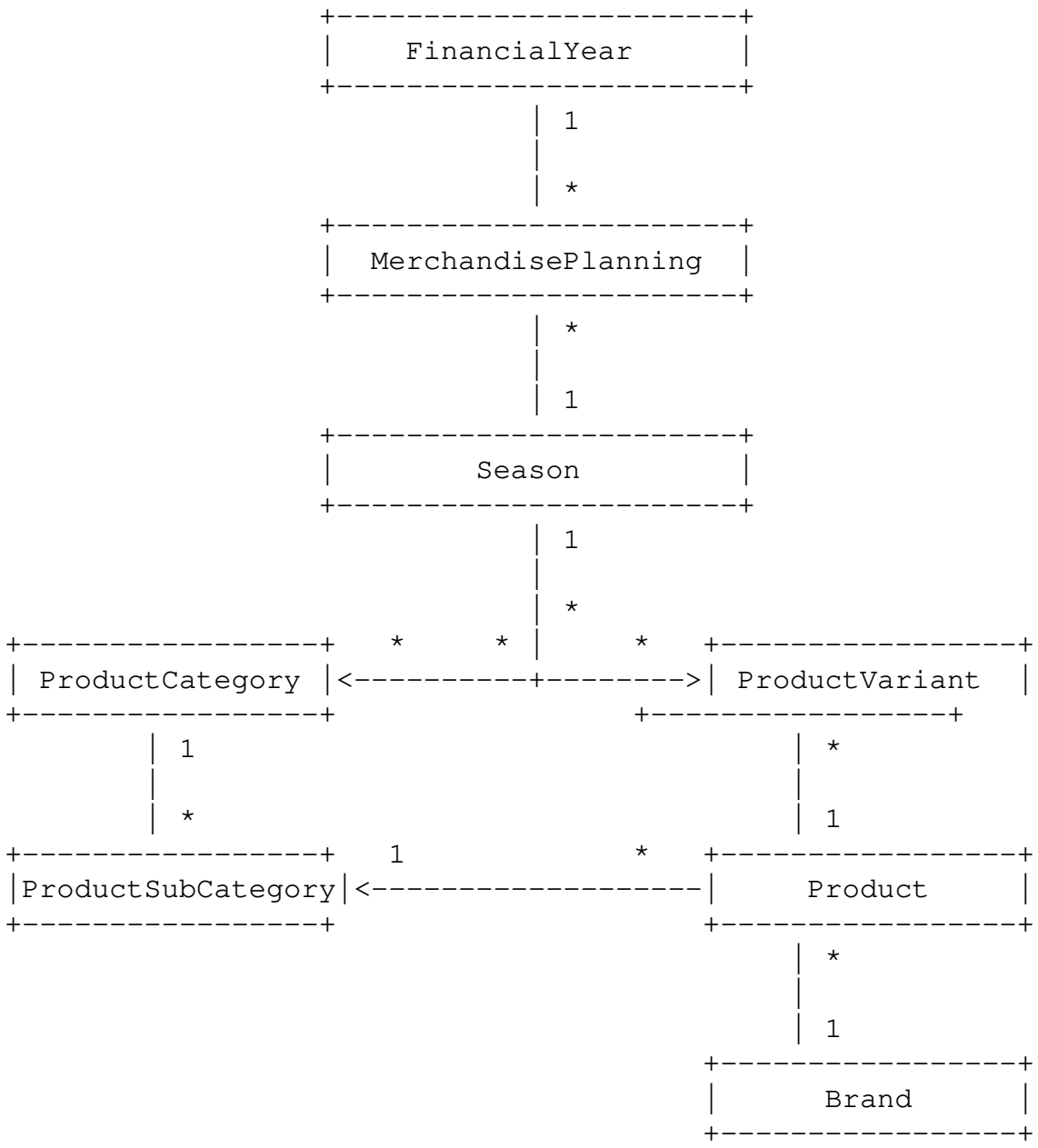
```
[ FinancialYear ]  <--- (financial_year_id) ---  [ MerchandisePlanning
```

1. **`MerchandisePlanning` -> `FinancialYear` (`belongsTo`):**
    - **Foreign Key:** `financial_year_id`
    - **Purpose:** Maps the merchandise plan to a specific active financial year (e.g., `FY 2026-2027`).

2. **`MerchandisePlanning` -> `Season` (`belongsTo`):**
    - **Foreign Key:** `season_id`
    - **Purpose:** Associates the sales targets and procurement budgets with a specific retail season (e.g., *Diwali*, *Winter Collection*).

2.2 Extended Product Information Management (PIM) Model Mapping

The `MerchandisePlanning` module interacts with the broader Product Information Management (PIM) catalog ecosystem:



Detailed Model Interconnections:

- **`ProductCategory` & `ProductSubCategory`:** Organize items into hierarchical merchandise divisions (e.g., `Apparel` -> `Men's Wear` -> `Formal Shirts`).
- **`Brand`:** Links master product records to vendor manufacturers (`Royal Silk Mills`).
- **`Product`:** Base product master containing default units, HSN codes, and base prices.
- **`ProductVariant`:** Connects `Season`, `Color`, `Size`, `FabricType`, `Design`, and `Style` to specific SKU items.
- **`PriceList`:** Controls wholesale and retail pricing tiers for variants during planned seasonal dates.

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3. Database Schema Overview

Table: `merchandise_plannings`

Column Name	Data Type	Key Type	Description

`id`	BigInteger (Auto-Increment)	Primary Key	Unique record identifier
`financial_year_id`	ForeignId	Foreign Key	References `id` on `financial_years`
`season_id`	ForeignId	Foreign Key	References `id` on `seasons`
`target_sales`	Decimal(15, 2)	Attribute	Targeted sales revenue in INR
`planned_budget`	Decimal(15, 2)	Attribute	Allocated inventory purchase budget
`created_at`	Timestamp	Standard	Record creation timestamp
`updated_at`	Timestamp	Standard	Record last modification timestamp

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4. Current Implementation vs Recommended Enhancements

Current Architecture State:

The current `MerchandisePlanning` model functions as a **macro-level financial forecasting tool**. It tracks total target sales and overall planned budgets per season.

Recommended Architectural Enhancements:

For enterprise ERP deployments requiring granular category-level or branch-level planning, the schema can be extended as follows:

1. **Branch-Level Allocation (`branch_id`):**
    - Add `foreignId('branch_id')->nullable()->constrained('branches')` to `merchandise_plannings`.
    - **Benefit:** Allows branch managers to set distinct seasonal sales targets for `Coimbatore Branch` vs `Chennai Branch`.
2. **Category-Level Budgeting (`product_category_id`):**
    - Add `foreignId('product_category_id')->nullable()->constrained('product_categories')`.
    - **Benefit:** Allows merchandise planners to allocate specific procurement budgets per category (e.g., allocating ₹ 25,00,000 for *Silk Sarees* and ₹ 15,00,000 for *Men's Wear*).